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Prof. Dr. Michael Walter,
Chairman of Quantum Information,
Faculty of Computer Science & Cluster of Excellence CASA,
Ruhr University Bochum.

Dear Michael Walter,

I am writing to express my interest in joining your research group as a post-doctoral student for the open quantum information position. I am currently a PhD Candidate working under the supervision of Rob W. Spekkens on topics in quantum information and quantum foundations at the Perimeter Institute for Theoretical Physics in Canada. I will be graduating April 2023.

In my latest paper, which was heavily inspired by the final chapter of your PhD thesis (Walter 2014), I developed an asymptotic solution to the quantum marginal problem (QMP) for overlapping marginals (Fraser 2022).

While my initial motivation for tackling the QMP originates from promising connections to my earlier research on quantum (and classical) causal networks (Fraser and Wolfe 2018; Fraser 2020) and multipartite entanglement (Schmid et al. 2020), I have since fallen in love with the rich cross-section of mathematical tools required to articulate its solution.

I am currently seeking opportunities to work on challenging problems in quantum information theory pertaining to the structure of multipartite quantum states using methods from representation theory, invariant theory, diagrammatic algebras, large deviation theory and convex geometry. While I am self-driven, I am also eager to collaborate with others.

I look forward to discussing potential research projects with you, but for now, here are a few topics I wish explore further:

- The link between the unipartite QMP and the asymptotics of Kronecker coefficients is well-known (Klyachko 2006; Christandl and Mitchison 2006). The general solution for the QMP outlined in my latest paper (Fraser 2022), when

restricted to the special case of unipartite marginals, yields quantities which asymptotically *approximate* Kronecker coefficients. Using techniques from geometric group theory, I'd like to explicitly bound the error of these approximations and analyze their limit as $n \to \infty$.

- The correspondence between quantum entropies and the recoupling theory of the symmetric group outlined in (Christandl, Şahinoğlu, and Walter 2018) can be extended to higher-order representation-theoretic identities. For instance, there exists an analogue of the Biedenharn-Elliott identity (which relates $6j$-symbols for $SU(2)$) for the recoupling coefficients of S_n. What do these identities, and their graphical symmetries, imply about the structure of multipartite quantum states and their entropies?
- Broadly and loosely speaking, the essential mathematical machinery underlying "random tensor networks" is to sample the components of a tensor network uniformly, and then calculate/bound the expected value of certain quantities of interest (moments, invariants, entropies, limits thereof, etc.) of the resulting quantum state. As you know, random tensor networks have proven to be an excellent toy model in the context of holographic duality for relating the geometric properties of a tensor network to the entropic and entanglement properties of the resulting multipartite quantum state (Hayden et al. 2016).
  The related field of quantum causal networks is interested in essentially the same correspondence between i) a given network or circuit of quantum states and channels and ii) the correlation properties of the multipartite quantum state that network generates (Navascues et al. 2020). Preliminary research suggests that by appropriately dilating the components of a quantum causal network and then uniformly sampling its unitary components, its seems possible to derive families of necessary conditions for the compatibility of a quantum state with a given causal network and furthermore prove their sufficiency.

Attached to this letter are i) a copy of my CV which includes a publication list, ii) contact details for two referees, and iii) a copy of my graduate and undergraduate transcript from the University of Waterloo.

Thank you for your time and consideration.

Sincerely,

T.C. Fraser

References

Walter, Michael (2014). "Multipartite quantum states and their marginals". In: *arXiv preprint arXiv:1410.6820*.

Fraser, Thomas C (2022). "A sufficient family of necessary inequalities for the compatibility of quantum marginals". In: *arXiv preprint arXiv:2211.00685*.

Fraser, Thomas C and Elie Wolfe (2018). "Causal compatibility inequalities admitting quantum violations in the triangle structure". In: *Physical Review A* 98.2, p. 022113.

Fraser, Thomas C (2020). "A combinatorial solution to causal compatibility". In: *Journal of Causal Inference* 8.1, pp. 22–53.

Schmid, David et al. (2020). "Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory". In: *arXiv preprint arXiv:2004.09194*.

Klyachko, Alexander A (2006). "Quantum marginal problem and N-representability". In: *Journal of Physics: Conference Series*. Vol. 36. 1. IOP Publishing, p. 72.

Christandl, Matthias and Graeme Mitchison (2006). "The spectra of quantum states and the Kronecker coefficients of the symmetric group". In: *Communications in mathematical physics* 261.3, pp. 789–797.

Christandl, Matthias, M Burak Şahinoğlu, and Michael Walter (2018). "Recoupling coefficients and quantum entropies". In: *Annales Henri Poincaré*. Vol. 19. 2. Springer, pp. 385–410.

Hayden, Patrick et al. (2016). "Holographic duality from random tensor networks". In: *Journal of High Energy Physics* 2016.11, pp. 1–56.

Navascues, Miguel et al. (2020). "Genuine network multipartite entanglement". In: *Physical Review Letters* 125.24, p. 240505.